

SAT 7

Reading & Writing Module 1

Question # ID

SAT7 RW 1.1 a07rw101

Botanist Al Kovalski has pointed out that maple trees already thrive in a wide variety of climates and thus may _____ changes in climate better than some other tree species do. The alterations maples may undergo in response to a changing climate are likely to be relatively small and easily achieved.

1 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) relocate from

(B) refer to

(C) originate from

(D) adapt to

Answer

D

SAT7 RW 1.2 a07rw102

The following text is from John Muir's 1913 autobiography *The Story of My Boyhood and Youth*. Muir describes being on a boat.

The water was so clear that it was almost invisible, and when we floated slowly out over the plants and fishes, we seemed to be miraculously sustained in the air while exploring a veritable fairyland.

2 Mark for Review

As used in the text, what does the word "clear" most nearly mean?

(A) Simple

(B) Understandable

(C) Obvious

(D) Transparent

D

SAT7 RW 1.3 a07rw103

The recently observed gamma ray burst GRB 230307A lasted for 200 seconds, _____ for a burst generated by the merger of neutron stars. Bursts caused by neutron mergers typically last fewer than 2 seconds.

3 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) a coincidence

(B) a reprieve

(C) an incident

(D) an oddity

D

SAT7 RW 1.4 a07rw104

In 1776, the United States sent Benjamin Franklin to France to try to win the country's support in the United States' fight for independence from Great Britain. Franklin was very popular in France. This _____ surely helped him to convince France to assist the United States.

4 Mark for Review

Which choice completes the text with the most logical and precise word or phrase?

(A) thoughtfulness

(B) esteem

(C) controversy

(D) sincerity

B

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SAT7 RW 1.5 a07rw105

In the 1950s, scientists didn't know much about the ocean floor. Many scientists at the time believed that the ocean floor was mostly flat. But geologist Marie Tharp and her research partner, Bruce Heezen, proved that this idea was wrong. Using sonar data collected from the Atlantic Ocean, Tharp and Heezen showed that the floor was filled with canyons, mountains, and valleys.

5

Mark for Review

ABC

Which choice best describes the function of the underlined sentence in the text as a whole?

- (A) It identifies a scientific belief that Tharp and Heezen showed to be wrong.
- (B) It describes the design of Tharp and Heezen's experiment.
- (C) It emphasizes a disagreement between Tharp and Heezen.
- (D) It presents data to support a claim that Tharp and Heezen made.

Answer

A

SAT7 RW 1.6 a07rw106

In the early days of television in the 1940s, many people thought that US television programs would rely on the financial support of ad agencies and commercial sponsors, much like radio did. But advertisers hesitated to jump into a new space, particularly at a time when the manufacturing of new television sets was stalled due to the US's involvement in World War II. Broadcasters, like the National Broadcasting Company (NBC), needed to persuade advertisers to support their programming despite not knowing whether there would be a robust television audience to begin with.

6

Mark for Review

ABC

Which choice best describes the function of the underlined phrase in the text as a whole?

- (A) It compares the beginnings of radio programming with the beginnings of television programming in the United States.
- (B) It identifies a specific reason behind some advertisers' hesitation to support television.
- (C) It describes how broadcasters attempted to convince advertisers to support television.
- (D) It explains why a type of television programming was popular at the time.

B

SAT7 RW 1.7 a07rw107

Pteropods are small swimming snails with thin, delicate calcium carbonate shells. These animals are thought to be especially vulnerable to ocean acidification due to calcium carbonate's susceptibility to dissolution at lower pH values. Victoria L. Peck and colleagues recently found that the periostracum (a protective coating on pteropods' outer shells) prevents this dissolution when intact. Moreover, the team was surprised to discover that even when the periostracum is breached, pteropods can still mitigate damage by rebuilding the inner shell wall.

7

Mark for Review

ABC

Which choice best describes the main purpose of the text?

- (A) To call for additional research on biological mechanisms that improve pteropod survival rates
- (B) To discuss a conclusion drawn in a study of calcium carbonate's role in protecting the periostracum of pteropods
- (C) To address some of the ways ocean acidification has altered pteropod behavior over time
- (D) To present findings that suggest that a concern about the effects of ocean acidification on pteropod shells may be unwarranted

D

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SAT7 RW 1.8 a07rw108

Answer

D

Text 1

Little is known about how plate tectonics—wherein slabs of Earth's crust move over, under, away from, and against one another—began. Some researchers contend that tectonic movements began around 3 billion years ago, often noting that computer models of Earth's mantle temperature at the time indicate that the mantle would have been sufficiently molten to enable the plates to move.

Text 2

Ultimately, any plausible claim about the inception of tectonic movement must rest on empirical evidence from the geological record. Researcher Wriju Chowdhury and his team analyzed the geochemistry of zircon crystals to gain insight into the chemical composition of the magma from which the crystals formed and, based on the data, compellingly argue that plate tectonics may have been occurring as early as 4.2 billion years ago.

8

Mark for Review

ABC

Based on the texts, how would the author of Text 2 most likely respond to what "some researchers contend" as described in Text 1?

- (A) By suggesting that the temperature of Earth's mantle 3 billion years ago was likely insufficient to allow for the level of tectonic movement predicted by computer models
- (B) By distinguishing between computer models of Earth's mantle temperature that reliably predict the onset of plate tectonics and those that do not
- (C) By indicating that computer models of Earth's mantle temperature are still being improved such that new models tend to be much more reliable than their predecessors
- (D) By asserting that a more definitive form of evidence than the computer models suggests a different timeline for the onset of plate tectonics on Earth

SAT7 RW 1.9 a07rw109

In 2018, scientists discovered an immense aggregation of *Muusoctopus robustus* (pearl octopuses) along a hydrothermal vent 3,200 meters beneath the ocean's surface. Water temperatures at this site—named the Octopus Garden—climb as high as 11°C, much warmer than the ambient 1.6°C typical at this depth. Based on observations made over three years, scientists concluded that temperatures at the site likely confer reproductive benefits and that the site is used exclusively for reproduction—6,000 *M. robustus* adults, hatchlings, and eggs were observed at the garden, but no juveniles were present.

9

Mark for Review

ABC

Which statement about *M. robustus* and the Octopus Garden is best supported by the text?

- (A) *M. robustus* leave the Octopus Garden upon reaching an intermediary stage of development.
- (B) The *M. robustus* population at the Octopus Garden remains stable despite variations in water temperature.
- (C) *M. robustus* nests in the Octopus Garden contain on average fewer but larger eggs than nests at similar ocean depths.
- (D) The Octopus Garden provides an ideal feeding ground for *M. robustus* hatchlings.

A

SAT7 RW 1.10 a07rw110

Conservationists worldwide are working to protect ecosystems from habitat destruction and biodiversity loss, and in many cases, initiatives that rely on natural features or processes can help address such challenges. In response to a rapidly dwindling population of blueback salmon, the Quinault Indian Nation (a tribe in Washington State) partnered with the conservation organization Wild Salmon Center to restore naturally occurring logjams in the Quinault River. The logjams create shady pools where the blueback salmon can rest and spawn, thus promoting blueback population recovery.

10

Mark for Review

ABC

Which choice best states the main idea of the text?

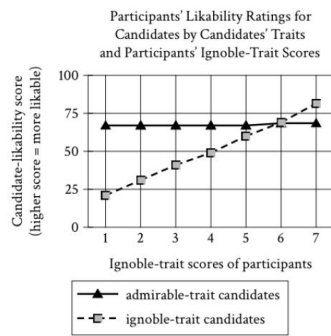
- (A) A partnership between the Quinault Indian Nation and Wild Salmon Center shows the importance of collaborative approaches to preserving biodiversity.
- (B) Nature-based approaches can be effective ways to achieve conservation goals.
- (C) As indicated by a recent project, logjams help the blueback salmon thrive and reproduce.
- (D) Scientists now realize that nature-based conservation methods offer better long-term solutions to environmental issues than methods that are not nature-based do.

B

Question # ID
SAT7 RW 1.11 a07rw111

Answer

B



Alessandro Nai et al. presented study participants with vignettes about fictive

political candidates, portraying them as embodying a personality trait widely considered admirable (e.g., agreeableness) or one considered ignoble (e.g., cynicism). A survey recorded participants' ratings of the candidates' likability and showed that across participants, ignoble-trait candidates were less likable than admirable-trait candidates. However, when the researchers factored in the participants' own personality-trait scores, on a scale of 1 (least ignoble) to 7 (most ignoble), they concluded that this relative ranking of candidates persisted except among the participants with high ignobility scores.

11 Mark for Review

Which choice best describes data from the graph that support the researchers' conclusion?

- (A) There was a strong positive correlation between participants' ignobility scores and admirable-trait candidates' likability ratings, but there was no correlation between ignobility scores and ignoble-trait candidates' likability ratings.
- (B) Participants with an ignobility score of 5 or less rated admirable-trait candidates as more likable than ignoble-trait candidates, whereas participants with an ignobility score of 6 or more rated ignoble-trait candidates as equally likable as or even more likable than admirable-trait candidates.
- (C) Overall, participants rated admirable-trait candidates as quite likable, and that rating was not significantly affected by the participants' ignobility scores.
- (D) Unlike participants with an ignobility score of 6, participants with an ignobility score either greater or less than 6 gave admirable-trait candidates and ignoble-trait candidates different likability ratings.

SAT7 RW 1.12 a07rw112

Puerto Rico is an island in the Caribbean Sea. Indigenous people there started raising guinea pigs about 1,700 years ago. Guinea pigs had originally been domesticated much earlier in both Colombia and Peru. So were guinea pigs brought to Puerto Rico from Colombia or from Peru? Ancient Caribbean trade routes connected Puerto Rico with Colombia but not with Peru. Therefore, guinea pigs in Puerto Rico probably came from Colombia and descended from Colombian guinea pigs.

12 Mark for Review

Which finding, if true, would most directly weaken the underlined claim?

- (A) Ancient guinea pigs in Puerto Rico were genetically less similar to ancient guinea pigs in Colombia than to ancient guinea pigs in Peru.
- (B) Guinea pigs are common in ancient Puerto Rican art, especially in pottery.
- (C) Modern breeds of guinea pigs don't look like images of guinea pigs in ancient art from Puerto Rico, Colombia, and Peru.
- (D) The guinea pig population of ancient Colombia was much larger than the guinea pig population of ancient Peru.

SAT7 RW 1.13 a07rw113

"Ad recall" measures how memorable an advertising campaign is. To provide advertisers with information about their ads' memorability, a social media site regularly surveys users about whether they remember ads they had recently interacted with on the site. In a study that drew on this survey data, advertising researcher Kristen Sussman and colleagues noted that different kinds of social media interactions involve different levels of cognitive engagement: commenting on or sharing a post is more cognitively demanding than is clicking on embedded links or on a "like" button. The researchers hypothesized that interactions indicating high levels of cognitive engagement with ad content would result in relatively high levels of ad recall.

13 Mark for Review

Which finding, if true, would most directly support the researchers' hypothesis?

- (A) Users who interacted with an ad were much more likely to do so by clicking on the ad's "like" button than they were to interact with the ad in any other way.
- (B) Users who interacted with an ad were significantly more likely to purchase the advertised product at the time they saw the ad than were users who saw the ad but did not interact with it.
- (C) Compared with users who clicked on links in an ad, users who commented on that same ad were significantly more likely to remember seeing the ad when surveyed two days later.
- (D) Although users who shared an ad were highly likely to remember details from the ad when surveyed two days later, those same users tended to forget those details when surveyed again a week later.

A

C

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Question # ID

SAT7 RW 1.14 a07rw114

Answer

A

Properties of Select Rotating Radio Transients

Name	Right ascension (hours)	Period (seconds)	Frequency (hertz)
J0545-03	5:45	1.074	0.931
J1654-2335	16:54:03	0.545	1.834
J0103+54	1:03:37	0.354	2.822
J0121+53	1:21	2.725	0.367
J0614-03	6:15	0.136	7.353

A student is researching rotating radio transients (RRATs), a subclass of pulsar stars characterized by short pulses of radio waves. The time between consecutive pulses of an RRAT is referred to as a period. Looking at the table, the student determines that _____

14 Mark for Review

ABC

Which choice most effectively uses data from the table to complete the statement?

- (A) J0614-03 has the shortest amount of time between consecutive pulses of all the RRATs in the table.
- (B) J0545-03 and J0121+53 have the same amount of time between consecutive pulses.
- (C) J1654-2335 has the longest amount of time between consecutive pulses of all the RRATs in the table.
- (D) J0103+54 and J0121+53 both have more than one second of time between consecutive pulses.

SAT7 RW 1.15 a07rw115

Narwhals are shy whales that live in the remote Arctic Ocean. Some of them have a long tusk, like a unicorn horn, with sensitive nerves. Narwhals are known for this tusk, but many actually don't have one and its purpose is unknown. One group of scientists came up with a possible purpose in 2014. The scientists suggested that the tusk may help narwhals determine when water around them is likely to start freezing and become dangerous for them. Marine biologist Kristin Laidre disagrees with that idea, though. She reasons that if the narwhal's tusk serves such an important purpose, then it's most likely that _____

15 Mark for Review

ABC

Which choice most logically completes the text?

- (A) some narwhals would seek a new habitat.
- (B) fewer marine animals would also have tusks.
- (C) more narwhals would have a tusk.
- (D) narwhals would become less shy over time.

C

SAT7 RW 1.16 a07rw116

Zydeco music originated in the French Creole community of southwest Louisiana. One instrument that gives zydeco its unique sound is the vest frottoir. The vest frottoir _____ a wearable washboard that is played by rubbing spoons or bottle openers against it.

16 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

- (A) have been
- (B) is
- (C) were
- (D) are

B

SAT7 RW 1.17 a07rw117

Featuring jagged peaks of black ink surrounded by hazy swirls of blue and green paint, Zhang Daqian's 1983 painting *Panorama of Mount Lu* is inspired by the tradition of *qingli shanshui*, a type of Chinese landscape painting _____ by the use of blue and green hues to depict ethereal, otherworldly landscapes.

17 Mark for Review

ABC

Which choice completes the text so that it conforms to the conventions of Standard English?

- (A) has been characterized
- (B) will be characterized
- (C) characterized
- (D) is characterized

C

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Answer

Question # ID

SAT7 RW 1.18 a07rw118

In his *Naturalis historia*, Pliny the Elder praised Hipparchus's star catalog, a second-century BCE list of roughly 850 different stars' celestial positions. For centuries, scholars dreamed about locating a copy of this legendary lost ____ fantasy (partially) became reality in 2022, when researchers uncovered traces of the star catalog on a palimpsest, a reused parchment.

18 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

- (A) work, that
- (B) work that
- (C) work. That
- (D) work and that

C

SAT7 RW 1.19 a07rw119

With a blend of traditional design elements, such as arched Gothic ceilings, and modern ones, such as floor-to-ceiling ____ design splits the difference between old and new, a mixture that is increasingly seen in home interiors in the US.

19 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

- (A) windows; transitional
- (B) windows—transitional
- (C) windows. Transitional
- (D) windows, transitional

D

SAT7 RW 1.20 a07rw120

When a given industry—water and electricity are two well-known examples—carries high infrastructural start-up costs and other barriers that discourage competition, ____ of just one or two suppliers per municipality. Such industries are known as natural monopolies.

20 Mark for Review

Which choice completes the text so that it conforms to the conventions of Standard English?

- (A) these often consist
- (B) they often consist
- (C) it often consists
- (D) this often consists

C

SAT7 RW 1.21 a07rw121

Famous for its four-degree tilt, the leaning Garisenda Tower is a popular attraction in Bologna's city center. However, measurements taken in 2023 showed that the tower was rotating in a concerning way. ____ city officials closed the area around the tower so experts could explore solutions to stabilize the historical twelfth-century structure.

21 Mark for Review

Which choice completes the text with the most logical transition?

- (A) Similarly,
- (B) As a result,
- (C) For example,
- (D) In comparison,

B

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Answer

Question # ID
SAT7 RW 1.22 a07rw122

Guard cells are specialized cells that are part of a plant's pores. These cells help regulate the amount of carbon dioxide a plant takes in. ____ they help regulate a plant's water loss.

22 Mark for Review

Which choice completes the text with the most logical transition?

- (A) Additionally,
- (B) Previously,
- (C) In conclusion,
- (D) Instead,

A

SAT7 RW 1.23 a07rw123

Marcel Duchamp intended his 1917 so-called ready-made sculpture *Fountain* to challenge then-prevailing conceptions about the nature of art. ____ Duchamp's *Fountain* did just that, raising the question of whether displaying any object in an art gallery could be said to transform the object—even, as Duchamp's sculpture was, a urinal—into a legitimate work of art.

23 Mark for Review

Which choice completes the text with the most logical transition?

- (A) Similarly,
- (B) Indeed,
- (C) Instead,
- (D) In addition,

B

SAT7 RW 1.24 a07rw124

In 2021, a model developed by astrophysicist Catherine Zucker and her research team revealed that the same supernovas responsible for the creation and ongoing expansion of the Local Bubble—a 14-million-year-old cavity in the Milky Way—are likely responsible for the formation of new stars. ____ this model detailed how the bubble's expansion trapped interstellar clouds of gas and dust that became stars upon their eventual collapse.

24 Mark for Review

Which choice completes the text with the most logical transition?

- (A) Hence,
- (B) However,
- (C) Admittedly,
- (D) Specifically,

D

SAT7 RW 1.25 a07rw125

While researching a topic, a student has taken the following notes:

- Musicians around the world have used protest songs to raise awareness about human rights violations.
- US folk singer Aunt Molly Jackson released the protest song "Poor Miner's Farewell" in 1932.
- It exposed the unlivable wages and dangerous working conditions coal miners faced in Kentucky during the 1920s and 1930s.
- South African singer-songwriter Hugh Masekela released the protest song "Bring Him Back Home" in 1987.
- It called on the South African government to free Nelson Mandela, an anti-apartheid leader who'd been unjustly imprisoned.

25 Mark for Review

The student wants to contrast the song "Poor Miner's Farewell" with the song "Bring Him Back Home." Which choice most effectively uses relevant information from the notes to accomplish this goal?

- (A) The songs "Poor Miner's Farewell" and "Bring Him Back Home" both raised awareness about human rights violations.
- (B) While both are protest songs, "Poor Miner's Farewell" is about coal miners in Kentucky, whereas "Bring Him Back Home" is about the anti-apartheid leader Nelson Mandela.
- (C) Hugh Masekela's song "Bring Him Back Home," released in 1987, called on the South African government to free Nelson Mandela.
- (D) Released in 1932 by Aunt Molly Jackson, the song "Poor Miner's Farewell" was a protest against the unlivable wages and dangerous working conditions faced by Kentucky coal miners.

B

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Answer

Question # ID

SAT7 RW 1.26 a07rw126

While researching a topic, a student has taken the following notes:

- Tibetan mastiffs are large dogs native to the Himalayas.
- A mutation in their EPAS1 gene prevents excess hemoglobin production.
- A mutation in their HBB gene boosts hemoglobin's oxygen-carrying ability.
- These mutations enable the dogs to withstand hypoxic (low-oxygen) conditions at high altitudes.
- In a 2016 study, Zhen Wang and colleagues noted that Tibetan wolves' DNA has the same EPAS1 and HBB mutations.
- Wang and colleagues determined that the dogs first acquired these mutations by interbreeding with Tibetan wolves around 24,000 years ago.

26

Mark for Review

ABC

The student wants to present the conclusion of Zhen Wang and colleagues' 2016 study. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- (A) Like Tibetan mastiffs, Tibetan wolves can withstand hypoxic conditions at high altitudes.
- (B) Both Tibetan mastiffs and Tibetan wolves have mutations in their EPAS1 and HBB genes, which prevent excess hemoglobin production and boost hemoglobin's oxygen-carrying ability, respectively.
- (C) In addition to preventing excess hemoglobin production, a mutation in Tibetan mastiffs' HBB gene boosts hemoglobin's oxygen-carrying ability.
- (D) By interbreeding with Tibetan wolves around 24,000 years ago, Tibetan mastiffs acquired the genetic mutations that enable them to withstand hypoxic conditions.

D

SAT7 RW 1.27 a07rw127

While researching a topic, a student has taken the following notes:

- Bike-share programs provide bicycles for shared use.
- In docked bike sharing, riders rent a bike and return it to designated docking stations.
- Docked programs are orderly and offer consistency to riders but require significant space and money to implement.
- In dockless bike sharing, riders locate a bike and leave it wherever they choose.
- Dockless programs are relatively simple and inexpensive to implement and offer flexibility to riders.
- Dockless programs can be disorganized.

27

Mark for Review

ABC

The student wants to compare some disadvantages of docked and dockless bike-share programs. Which choice most effectively uses relevant information from the notes to accomplish this goal?

- (A) Dockless programs can be disorganized; docked programs, on the other hand, offer order and consistency.
- (B) Worth noting is that while dockless programs are relatively easy and inexpensive to implement, they are less flexible than docked programs.
- (C) Docked programs are more resource-intensive than dockless programs, but they avoid some of the latter's organizational challenges.
- (D) Though dockless programs offer flexibility, docked bike-share programs provide bicycles for shared use.

C